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Project Title:	Extinction and Instability Mechanisms of Polymerization Fronts		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
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Zip Code:	90089-1453	**Congressional District:**	33
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
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Recent advances in polymer chemistry have led to the development of monomers and initiation agents providing propagating polymerization fronts driven by the exothermicity of the polymerization reaction and the transport of heat from the polymerized product to the monomer. The purpose of this study is to determine the mechanisms of extinction and instability and thereby determine means to obtain more useful product material at earth gravity and µg. Experiments are being conducted in two distinct geometries, specifically Hele-Shaw cells and round tubes. Our prior experiments in gas combustion in round tubes of varying diameter have demonstrated two distinct extinction limits due to these processes; it will be determined if the same applies to polymer fronts. In addition to µg experiments, in the Hele-Shaw cell varying the angle of the cell relative to the horizontal will be used to assess buoyancy effects _ though with limitations because a minimum cell thickness and thus minimum Rayleigh number is required to avoid heat-loss induced quenching. The experiments will measure propagation rates and front shapes using direct video, laser-induced fluorescence (LIF) imaging, a technique not previously used in studies of polymerization fronts, along with temperature

Task Description:	measurements using thermocouples and laser interferometry, and flow properties via particle tracking and laser Doppler velocimetry. The polymerized products will be characterized using NMR to determine the conversion efficiency and Gel Permeation Chromatography to measure the product molecular weight distribution. Also, prior experiments in aqueous autocatalytic chemical reaction fronts in Hele-Shaw cells have shown a previously unknown manifestation of the Saffman-Taylor instability due to surface tension effects, even though the reactants and products are completely miscible in each other. We are determining if such surface tension effects exist in polymer fronts as well. A companion numerical modeling effort is being performed. A novel aspect of these studies is the inclusion of Korteweg stresses that enable modeling of surface tension-like forces in regions of steep concentration or temperature gradients at the interface between miscible fluids. Another important aspect of the modeling study is the inclusion of heat losses to walls, which are anecdotally know to cause extinction of polymer fronts but have not been the subject of systematic evaluation in prior literature. The use of polymerization processes based on this mode of polymerization has many applications including rapid curing of polymers without external heating, uniform curing of thick samples, solventless preparation of some polymers, and filling/sealing of structures having cavities of arbitrary shape without having to heat the structure externally. One important limitation of this process is that the fronts extinguish when they try to propagate through channels that are too narrow (probably due to conductive heat losses) or too wide (for unknown reasons, which we propose to be convective heat losses driven by buoyancy-induced flow.) Even when extinction does not occur, convective and buoyant instabilities can affect the structure and properties of the resulting polymerized materials as well as the propagation rates of the fronts. Microgravity experiments are an indispensable aspect of this project because polymer fronts are very likely subject to quenching by buoyancy-driven mechanisms, and buoyancy also leads to a wide variety of instabilities, which may overwhelm effects due to surface tension, viscosity difference, and thermal expansion/contraction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The use of polymerization processes based on this mode of polymerization has many applications including rapid curing of polymers without external heating, uniform curing of thick samples, solventless preparation of some polymers, and filling/sealing of structures having cavities of arbitrary shape without having to heat the structure externally.
Task Progress:	Frontal polymerization at ambient pressure using water as the solvent were demonstrated by using a very reactive monomer (AA) and free-radical initiator (AP) that enabled self-propagating free-radical polymerization fronts to occur at temperatures below 100°C. The range of AA and AP concentrations producing polymerization fronts is limited by extinguishments via heat losses at low AA and AP concentrations, and boiling at high concentrations. The heat losses leading to extinction appear to be due to buoyant convection from the tube wall to ambient air. When the wall condition is isothermal, the range of compositions exhibiting propagating fronts is much smaller and no AA and AP concentrations exhibited non-bubbling fronts. It is shown, however, that with sufficiently large tubes, non-bubbling propagating fronts could probably be obtained. It is suggested that frontal polymerization in water at ambient pressure could lead to new practical applications because of the cost and environmental issues associated with solvents other than water and the inconvenience of using pressures high than atmospheric to avoid boiling.
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